

EXPERIMENT 3

EXPLORATORY DATA ANALYSIS (EDA) – DATA CLEANING AND DATA TRANSFORMATION

Aim

To identify and handle missing values, duplicate records, outliers, inconsistent data, and perform data transformation techniques such as normalization and encoding using Python Pandas.

Algorithm 1

Identifying Missing Values

Step 1

Import pandas library

Step 2

Load dataset

Step 3

Use isnull() function

Step 4

Count missing values

Step 5

Display result

Program :

```
[1]: import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler, LabelEncoder

[2]: df = pd.read_csv("healthcare-dataset-stroke-data.csv")

print(df.head())
```

	id	gender	age	hypertension	heart_disease	ever_married	\
0	9046	Male	67.0	0	1	Yes	
1	51676	Female	61.0	0	0	Yes	
2	31112	Male	80.0	0	1	Yes	
3	60182	Female	49.0	0	0	Yes	
4	1665	Female	79.0	1	0	Yes	

	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	\
0	Private	Urban	228.69	36.6	formerly smoked	
1	Self-employed	Rural	202.21	NaN	never smoked	
2	Private	Rural	105.92	32.5	never smoked	
3	Private	Urban	171.23	34.4	smokes	
4	Self-employed	Rural	174.12	24.0	never smoked	

	stroke
0	1
1	1
2	1
3	1
4	1

Program 2

Count Missing Values

```
[4]: print(df.isnull())
```

	id	gender	age	hypertension	heart_disease	ever_married	\
0	False	False	False	False	False	False	
1	False	False	False	False	False	False	
2	False	False	False	False	False	False	
3	False	False	False	False	False	False	
4	False	False	False	False	False	False	
...	...	...	...	...	...	...	
5105	False	False	False	False	False	False	
5106	False	False	False	False	False	False	
5107	False	False	False	False	False	False	
5108	False	False	False	False	False	False	
5109	False	False	False	False	False	False	

	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	\
0	False	False	False	False	False	
1	False	False	False	True	False	
2	False	False	False	False	False	
3	False	False	False	False	False	
4	False	False	False	False	False	
...	...	...	...	...	...	
5105	False	False	False	True	False	
5106	False	False	False	False	False	
5107	False	False	False	False	False	
5108	False	False	False	False	False	
5109	False	False	False	False	False	

	stroke
0	False
1	False
2	False
3	False
4	False
...	...
5105	False
5106	False

```
[5]: print(df.isnull().sum())
print("Total Missing Values:", df.isnull().sum().sum())

id          0
gender       0
age          0
hypertension 0
heart_disease 0
ever_married 0
work_type    0
Residence_type 0
avg_glucose_level 0
bmi         201
smoking_status 0
stroke       0
dtype: int64
Total Missing Values: 201
```

Algorithm 2

Replacing Missing Values

Step 1

Calculate mean value

Step 2

Replace null values

Step 3

Verify replacement

Program 3:

```
[7]: df["bmi"] = df["bmi"].fillna(df["bmi"].mean())
print(df)
```

	id	gender	age	hypertension	heart_disease	ever_married	bmi \
0	9046	Male	67.0	0	1	Yes	
1	51676	Female	61.0	0	0	Yes	
2	31112	Male	80.0	0	1	Yes	
3	60182	Female	49.0	0	0	Yes	
4	1665	Female	79.0	1	0	Yes	
...	...	...	...	...	...	...	...
5105	18234	Female	80.0	1	0	Yes	
5106	44873	Female	81.0	0	0	Yes	
5107	19723	Female	35.0	0	0	Yes	
5108	37544	Male	51.0	0	0	Yes	
5109	44679	Female	44.0	0	0	Yes	

	work_type	Residence_type	avg_glucose_level	bmi \
0	Private	Urban	228.69	36.600000
1	Self-employed	Rural	202.21	28.893237
2	Private	Rural	105.92	32.500000
3	Private	Urban	171.23	34.400000
4	Self-employed	Rural	174.12	24.000000
...	...	...	...	...
5105	Private	Urban	83.75	28.893237
5106	Self-employed	Urban	125.20	40.000000
5107	Self-employed	Rural	82.99	30.600000
5108	Private	Rural	166.29	25.600000
5109	Govt_job	Urban	85.28	26.200000

	smoking_status	stroke
0	formerly smoked	1
1	never smoked	1
2	never smoked	1
3	smokes	1
4	never smoked	1

Algorithm 3

Removing Duplicate Records

Step 1

Use duplicated()

Step 2

Count duplicate records

Step 3

Remove duplicates

Step 4

Verify dataset

Program 4

```
[8]: print(df.duplicated().sum())

df.drop_duplicates(inplace=True)

print(df.shape)

0
(5110, 12)
```

Algorithm 4

Detecting Outliers

Step 1

Calculate Quartiles

Step 2

Compute IQR

Step 3

Calculate lower and upper limits

Step 4

Identify outliers

Formula

Inter Quartile Range

$$IQR = Q_3 - Q_1$$

Outlier Limits

$$\text{Lower Limit} = Q_1 - 1.5 \times IQR$$

$$\text{Upper Limit} = Q_3 + 1.5 \times IQR$$

Program 5

```
Q1 = df["age"].quantile(0.25)
Q3 = df["age"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["age"] < lower) | (df["age"] > upper)]

print(outliers)
```

Empty DataFrame

Columns: [id, gender, age, hypertension, heart_disease, ever_married, work_type, Residence_type, avg_glucose_level, bmi, smoking_status, stroke]

Index: []

Visualization of Outlier

Employee Salary Analysis

Salary distribution showing potential Outlier.

Algorithm 5

Normalization

Purpose

Convert numerical values into a common scale.

Formula

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Program 6:

```
[5]: from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df[['bmi']] = scaler.fit_transform(df[['bmi']])

print(df[['bmi']])
```

	bmi
0	0.301260
1	NaN
2	0.254296
3	0.276060
4	0.156930
...	...
5105	NaN
5106	0.340206
5107	0.232532
5108	0.175258
5109	0.182131

[5110 rows x 1 columns]

Algorithm 6

Label Encoding

Purpose

Convert categorical data into numerical format.

Program 7

```
[13]: encoder = LabelEncoder()

df["gender"] = encoder.fit_transform(df["gender"])
df["ever_married"] = encoder.fit_transform(df["ever_married"])
df["work_type"] = encoder.fit_transform(df["work_type"])
df["Residence_type"] = encoder.fit_transform(df["Residence_type"])
df["smoking_status"] = encoder.fit_transform(df["smoking_status"])

print(df.head())
```

	id	gender	age	hypertension	heart_disease	ever_married	work_type	\
0	9046	1	67.0	0	1	1	2	
1	51676	0	61.0	0	0	1	3	
2	31112	1	80.0	0	1	1	2	
3	60182	0	49.0	0	0	1	2	
4	1665	0	79.0	1	0	1	3	

	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	1	228.69	0.301260	1	1
1	0	202.21	0.212981	2	1
2	0	105.92	0.254296	2	1
3	1	171.23	0.276060	3	1
4	0	174.12	0.156930	2	1

Program 8

One-Hot Encoding

```
[14]: encoded = pd.get_dummies(df["work_type"])

print(encoded.head())
```

	0	1	2	3	4
0	False	False	True	False	False
1	False	False	False	True	False
2	False	False	True	False	False
3	False	False	True	False	False
4	False	False	False	True	False

Data Transformation Summary

Technique	Purpose
Fill Missing Values	Handle null data
Drop Duplicates	Remove redundancy
Outlier Detection	Improve data quality
Normalization	Scale data
Label Encoding	Convert categories
One-Hot Encoding	Binary representation

Result

Thus, missing values, duplicate records, and outliers were identified and handled successfully. Data transformation techniques such as normalization and encoding were performed, and the dataset was prepared for further exploratory data analysis and machine learning applications.

